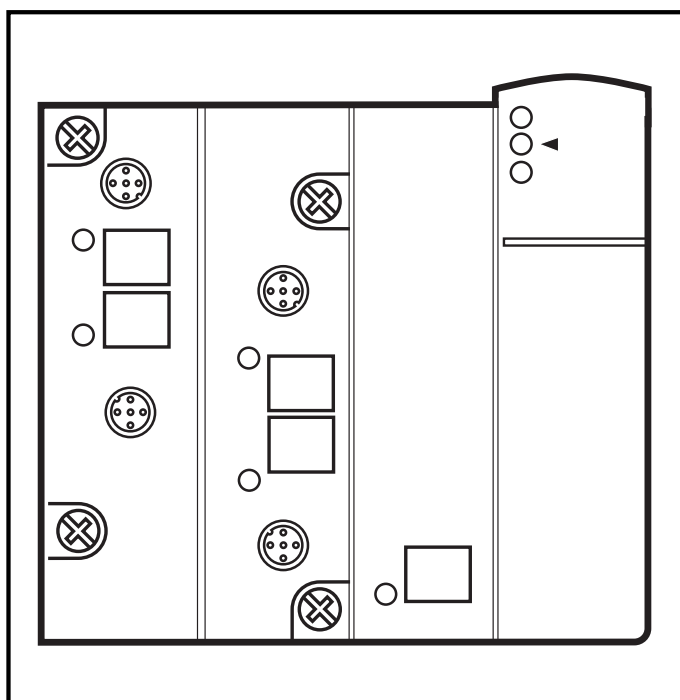


**Montageanleitung
Installation instructions
Notice de montage**

AS interface

**AS-i Modul
AS-i module
Module AS-i**

**AC2516
AC2517**



Function and features

The slave converts analogue input signals and transfers them to the AS-i master via the AS-Interface. The AS-i module operates as a slave with bidirectional data transfer in the AS-i network.

The data transfer to the host is asynchronous according to the AS-i profile S-7.3 and the AS-i specification V2.1.

- Current measurement 4 ... 20mA (AC2516)
or voltage measurement 0 ... 10V (AC2517)
- AS-i profile S-7.3.E
- The sensors are connected via M12 connection cables
- Maximum number of modules per AS-i system: 31
- R_i voltage measurement $> 100k\Omega$; R_i current measurement $< 50\Omega$
- Time for converting the measured values in the slave
 - for one channel: 20ms
 - for two channels: 120ms
 - for three channels: 180ms
 - for four channels: 240ms
- Sensor supply from AS-i (max. 100mA) or external 24V PELV voltage source (black flat cable, max. 500 mA)
- Resolution: 1 μ A (AC2516) or 1mV (AC2517)

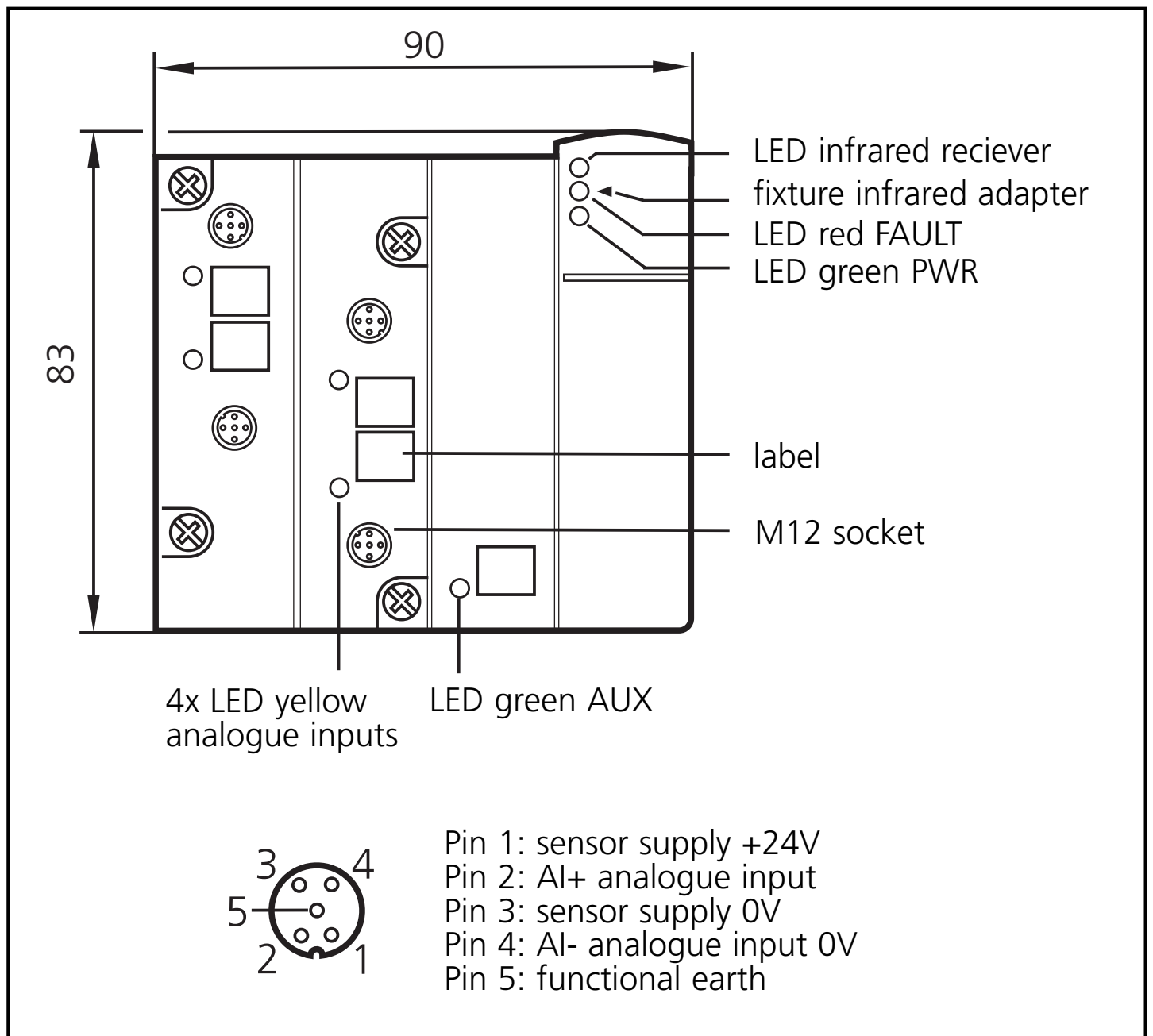
Installation / Addressing

When you use module lower parts without addressing socket (AC5000, AC5001 or AC5003) first address the module by placing it onto an addressing unit (AC1144) and assign a free address between 1 and 31.

When you use module lower parts with an addressing socket (AC5010 or AC5011) the modules can be addressed with the addressing adapter E70213 later on.

Mount the module onto the wired module lower part of the AS-i network, tightening torque 0.8Nm.

As an alternative FC/FC-E lower parts with earthing lead (AC5020...AC5023) can be used to increase the noise immunity.



Electrical connection

The analogue module is connected to the AS-Interface via the standardised EMS (supply from AS-i) or the E-EMS (supply from an ext. 24V PELV voltage source).

If the supply is to be from an external 24V source, a FC-E lower part (art. no. AC5003 or AC5011) must be used.



If a total of over 100mA is needed for the sensor supply, the supply must be from an external 24V PELV voltage source.

The supply is automatically changed when the external 24V voltage is supplied.



Risk of destruction:

When a combined sensor (pin 2: analogue output, pin 4: 24V output) is connected ensure that the switching output cannot switch. To do so, set the combined sensor accordingly (e.g. by selection of a switch point which cannot be reached or by the configuration "NPN switching").

Current measurement AC2516

For all the following wiring diagrams the indicated pin connection refers to the analogue module.

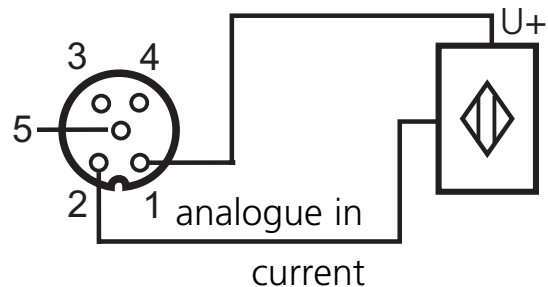
When an external link between pin 3 and pin 4 is used, the internal link can be deactivated by resetting the parameter bit P0!



The internal link (pin 3 and pin 4) must be activated via the parameter bit P0.

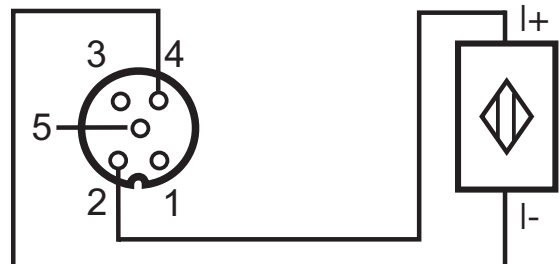
Connection of a 2-wire sensor without own supply

- Pin 1: sensor supply +24V
- Pin 2: AI+ analogue input
- Pin 3: sensor supply 0V
- Pin 4: AI- analogue input 0V
- Pin 5: functional earth



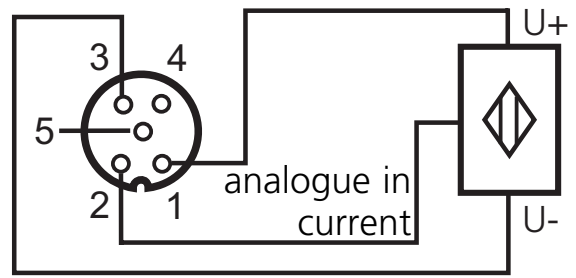
Connection of a 2-wire sensor with own supply

- Pin 1: sensor supply +24V
- Pin 2: AI+ analogue input
- Pin 3: sensor supply 0V
- Pin 4: AI- analogue input 0V
- Pin 5: functional earth



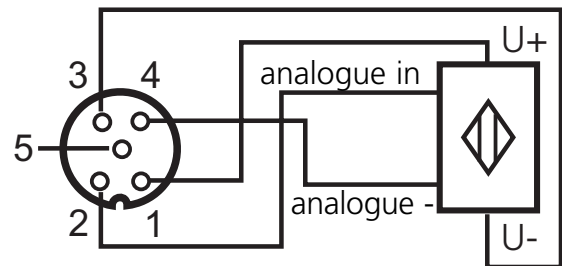
Connection of a 3-wire sensor without own supply

- Pin 1: sensor supply +24V
- Pin 2: AI+ analogue input
- Pin 3: sensor supply 0V
- Pin 4: AI- analogue input 0V
- Pin 5: functional earth



Connection of a 4-wire sensor without own supply

- Pin 1: sensor supply +24V
- Pin 2: AI+ analogue input
- Pin 3: sensor supply 0V
- Pin 4: AI- analogue input 0V
- Pin 5: functional earth



When a 4-wire sensor is connected the internal link between pin 3 and pin 4 must be deactivated by resetting the parameter bit P0.

Voltage measurement AC2517

 The parameter bit P0 is of no importance for the AC2517!

Connection of a 2-wire sensor without own supply

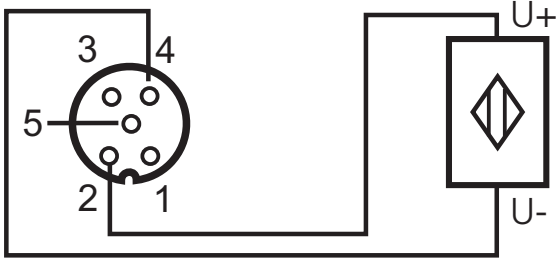
Pin 1: sensor supply +24V

Pin 2: AI+ analogue input

Pin 3: sensor supply 0V

Pin 4: AI- analogue input 0V

Pin 5: functional earth



Connection of a 3-wire sensor without own supply

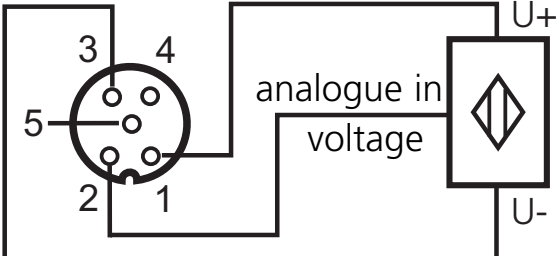
Pin 1: sensor supply +24V

Pin 2: AI+ analogue input

Pin 3: sensor supply 0V

Pin 4: AI- analogue input 0V

Pin 5: functional earth



 When a 3-wire sensor without own supply is connected the link between pin 3 and pin 4 has to be created externally!

Parameter setting of the analogue modules

Parameter bit	Designation	Description																														
P0 (not used for AC2517)	selection 2/3 wires / 4 wires	1 2-/ 3-wire operation (link active) 0 4-wire operation (link inactive)																														
P1, P2	channel activation	<table><tr><th>P1</th><th>P2</th><th>channel1</th><th>channel2</th><th>channel3</th><th>channel4</th></tr><tr><td>0</td><td>0</td><td>on</td><td>off</td><td>off</td><td>off</td></tr><tr><td>0</td><td>1</td><td>on</td><td>on</td><td>off</td><td>off</td></tr><tr><td>1</td><td>0</td><td>on</td><td>on</td><td>on</td><td>off</td></tr><tr><td>1</td><td>1</td><td>on</td><td>on</td><td>on</td><td>on</td></tr></table>	P1	P2	channel1	channel2	channel3	channel4	0	0	on	off	off	off	0	1	on	on	off	off	1	0	on	on	on	off	1	1	on	on	on	on
P1	P2	channel1	channel2	channel3	channel4																											
0	0	on	off	off	off																											
0	1	on	on	off	off																											
1	0	on	on	on	off																											
1	1	on	on	on	on																											
P3	periphery fault	1 periphery fault indication active 0 periphery fault indication not active																														

Operation AC2516

Check the safe functioning of the unit. Display by LEDs:

- LED yellow AI-1...AI off: no sensor connected
(at least one LED flashes because not all channels can be deactivated via the parameter bit P1/P2 (channel activation))
- LED yellow AI-1...AI-4 on: analogue signal in the measuring range
- LED yellow AI-1...AI-4 flashes: analogue signal outside the measuring - range or no sensor connected
- LED green PWR on: AS-i voltage is applied
- LED green AUX on: external 24V voltage is applied
- LED red FAULT on: AS-i communication error
- LED red FAULT flashes: periphery fault*

* periphery fault

A periphery fault is indicated:

- if at least one of the analogue signals is outside the value range.
- if nothing is connected to at least one analogue channel although the respective channel is activated

Operation AC2517

Check the safe functioning of the unit. Display by LEDs:

- LED yellow AI-1...AI off: no sensor connected
(at least one LED flashes because not all channels can be deactivated via the parameter bit P1/P2 (channel activation))
- LED yellow AI-1...AI-4 on: analogue signal in the measuring range or no sensor connected (it cannot be differentiated whether the 0V signal is applied or whether no sensor is connected)
- LED yellow AI-1...AI-4 flashes: analogue signal outside the measuring range (outside range)

- LED green PWR on: AS-i voltage is applied
- LED green AUX on: external 24V voltage is applied
- LED red FAULT on: AS-i communication error
- LED red FAULT flashes: periphery fault*

* periphery fault

A periphery fault is indicated if at least one of the analogue signals is outside the value range.

Measuring range of the analogue input modules

The measuring ranges, the states of the LEDs and their meaning are indicated in the following tables:

Analogue input module 4 ... 20mA - AC2516

Range 4 ... 20mA	Units dec.	Units hex.	LED AI1...AI4 analogue	Meaning
< 1mA	32767	7FFF	flashes	wire break
1mA ... 3.999mA	1000 ... 3999	03E8 ... 0F9F	on	below nominal range
4mA ... 20mA	4000 ... 20000	0FA0 ... 4E20	on	nominal range
20.001mA ... 23mA	20001 ... 23000	4E21 ... 59D8	on	above nominal range
> 23mA	32767	7FFF	flashes	outside range

Analogue input module 0 ... 10V - AC2517

Range 0 ... 10V	Units dec.	Units hex.	LED analogue	Meaning
0 ... 10V	0000 ... 10000	0000 ... 2710	on	nominal range
10.001V ... 11.5V	10001 ... 11500	2711 ... 2CEC	on	above nominal range
> 11.5V	32767	7FFF	flashes	outside range

Transmission time of the analogue values

The transmission time of the analogue values depends on the conversion time of the analogue signals into digital signals in the AS-i module and on the transmission time via the AS-Interface.

Example: Transmission time of 2 analogue channels

The conversion time for 2 analogue input signals into digital signals is 120ms. The transmission time of the 2 16-bit values via the AS-interface ideally is 7 AS-i cycles per value. For a cycle time of 5ms per AS-i cycle this results in a transmission time of $2 \times 7 \times 5\text{ms} = 70\text{ms}$ via the AS-Interface.

Thus the total transmission time for 2 analogue values ideally is 120ms (conversion time) + 70ms (transmission time) = 190ms.

Example: Transmission time of 4 analogue channels

The conversion time for 4 analogue input signals into digital signals is 240ms. The transmission time of the 4 16-bit values via the AS-interface ideally is 7 AS-i cycles per value. For a cycle time of 5ms per AS-i cycle this results in a transmission time of $4 \times 7 \times 5\text{ms} = 140\text{ms}$ via the AS-Interface.

Thus the total transmission time for 4 analogue values ideally is 240ms (conversion time) + 140ms (transmission time) = 380ms.